

NUMERICAL SOLUTIONS: LABORATORY REPORT

Topic: Simple Linear Regression Analysis (Least Squares Method)

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PART A. Data

Source & Full URL: World Bank Open Data (World Development Indicators)
URL: <https://data.worldbank.org/country/philippines>
Description: Real GDP vs Population of the Philippines (2005–2019).
Variables: Independent (x) = Population (Millions) | Dependent (y) = Real GDP (Billions USD, Constant 2015)

Year	Population (x) [Millions]	Real GDP (y) [Billions USD]
2005	84.1	181.2
2006	85.0	190.5
2007	86.0	200.7
2008	87.0	210.1
2009	88.0	215.4
2010	89.0	228.6
2011	90.1	240.2
2012	91.2	252.7
2013	92.3	265.9
2014	93.4	278.4
2015	94.6	292.1
2016	95.8	307.5
2017	97.0	324.8
2018	98.2	342.6
2019	99.4	361.8

PART B. Python Script

```
import os
import inspect
import matplotlib.pyplot as plt
import numpy as np

from reportlab.lib import colors
from reportlab.lib.pagesizes import letter
from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
from reportlab.platypus import (
    HRFloable,
    Image,
    Paragraph,
    Preformatted,
    SimpleDocTemplate,
    Spacer,
    Table,
    TableStyle,
)

# -----
# 1. DATASET & LEAST SQUARES CALCULATIONS
# -----
x = np.array([
    84.1, 85.0, 86.0, 87.0, 88.0,
    89.0, 90.1, 91.2, 92.3, 93.4,
    94.6, 95.8, 97.0, 98.2, 99.4
])

y = np.array([
    181.2, 190.5, 200.7, 210.1, 215.4,
    228.6, 240.2, 252.7, 265.9, 278.4,
    292.1, 307.5, 324.8, 342.6, 361.8
])

n = len(x)

sum_x = np.sum(x)
sum_y = np.sum(y)

sum_xy = np.sum(x * y)
sum_x2 = np.sum(x**2)

x_mean = np.mean(x)
y_mean = np.mean(y)

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - (sum_x)**2)
a0 = y_mean - a1 * x_mean

y_pred = a0 + a1 * x
residuals = y - y_pred

Sr = np.sum(residuals**2) # SSE
St = np.sum((y - y_mean)**2) # SST
r2 = (St - Sr) / St
sy_x = np.sqrt(Sr / (n - 2))

x_target = 105.0
y_predicted = a0 + a1 * x_target

# -----
# 2. GENERATE AND SAVE PLOTS
# -----
plt.figure(figsize=(5.5, 2.7))
plt.scatter(x, y, color='#1E3A2B', label='Data Points (2005-2019)',
            zorder=5, s=30)
plt.plot(x, y_pred, color='#2E5A44', linewidth=2, label=f'Fit: y = {a0:.2f} + {a1:.2f}x')
plt.title('Philippines Population vs. GDP', fontsize=9,
          fontweight='bold', color='#1E3A2B')
plt.xlabel('Population (Millions)', fontsize=7.5)
plt.ylabel('Real GDP (Billions USD)', fontsize=7.5)
plt.xticks(fontsize=7.5)
plt.yticks(fontsize=7.5)
plt.grid(True, linestyle='--', alpha=0.4, color='#A8DADC')
plt.legend(fontsize=7.5, frameon=True, facecolor='#F4F7F4')
plt.tight_layout()
plt.savefig('fit_plot.png', dpi=300)
plt.close()
```

```

plt.figure(figsize=(5.5, 2.7))
plt.scatter(x, residuals, color='#3B7A57', zorder=5, s=30)
plt.axhline(0, color='#1E3A2B', linestyle='--', linewidth=1)
plt.title('Residual Plot (y - y_pred)', fontsize=9,
fontweight='bold', color='#1E3A2B')
plt.xlabel('Population (Millions)', fontsize=7.5)
plt.ylabel('Residuals (Billions USD)', fontsize=7.5)
plt.xticks(fontsize=7.5)
plt.yticks(fontsize=7.5)
plt.grid(True, linestyle='--', alpha=0.4, color='#A8DADC')
plt.tight_layout()
plt.savefig('residual_plot.png', dpi=300)
plt.close()

# -----
# 3. REPORTLAB PDF BUILDING
# -----

pdf_file = "Linear_Regression_Report.pdf"
doc = SimpleDocTemplate(
    pdf_file,
    pagesize=letter,
    rightMargin=36,
    leftMargin=36,
    topMargin=24,
    bottomMargin=24
)

styles = getSampleStyleSheet()

C_DARK_GREEN = colors.HexColor('#1E3A2B')
C_MED_GREEN = colors.HexColor('#2E5A44')
C_ACCENT_MINT = colors.HexColor('#4E9F76')
C_PASTEL_BG = colors.HexColor('#E8F5E9')
C_CARD_BG = colors.HexColor('#F4F7F4')
C_BORDER = colors.HexColor('#C8E6C9')
C_TEXT = colors.HexColor('#2C3E35')

title_style = ParagraphStyle('DocTitle', parent=styles['Heading1'],
fontSize=16, leading=20, textColor=C_DARK_GREEN,
fontName='Helvetica-Bold')
h1_style = ParagraphStyle('PartHeader', parent=styles['Heading1'],
fontSize=13, leading=16, textColor=C_DARK_GREEN,
fontName='Helvetica-Bold', spaceBefore=8, spaceAfter=3)
h2_style = ParagraphStyle('SectionHeader',
parent=styles['Heading2'], fontSize=10, leading=13,
textColor=C_MED_GREEN, fontName='Helvetica-Bold', spaceBefore=5,
spaceAfter=3)
body_style = ParagraphStyle('BodyContent', parent=styles['Normal'],
fontSize=8, leading=11, textColor=C_TEXT, spaceAfter=3)
meta_style = ParagraphStyle('MetaContent', parent=styles['Normal'],
fontSize=8.5, leading=12, textColor=C_DARK_GREEN)
table_header_style = ParagraphStyle('TableHeaderContent',
parent=styles['Normal'], fontSize=8, leading=11,
textColor=colors.white, fontName='Helvetica-Bold')
code_style = ParagraphStyle('CodeStyle', parent=styles['Code'],
fontSize=6, leading=7.5, textColor=colors.HexColor('#A202C'),
fontName='Courier')

card_num_style = ParagraphStyle('CardNumStyle',
parent=styles['Normal'], fontSize=16, leading=18,
textColor=colors.white, fontName='Helvetica-Bold', alignment=1)
card_title_style = ParagraphStyle('CardTitleStyle',
parent=styles['Normal'], fontSize=8, leading=9,
textColor=colors.white, fontName='Helvetica-Bold', alignment=1)
card_text_style = ParagraphStyle('CardTextStyle',
parent=styles['Normal'], fontSize=7, leading=9, textColor=C_TEXT,
alignment=1)

elements = []

# Metadata Header
elements.append(Paragraph("NUMERICAL SOLUTIONS: LABORATORY REPORT",
title_style))
elements.append(Paragraph("<b>Topic:</b> Simple Linear Regression
Analysis (Least Squares Method)", body_style))
elements.append(Spacer(1, 3))

info_data = [
    Paragraph("<b>Name:</b> Villondo, Angel Rain L.", meta_style),
    Paragraph("<b>Section:</b> BSCE-3D", meta_style),
    Paragraph("<b>Date:</b> September 02, 2026", meta_style)
]
info_table = Table(info_data, colWidths=[200, 150, 190])
info_table.setStyle(TableStyle([
    ('BACKGROUND', (0, 0), (-1, -1), C_PASTEL_BG),
    ('PADDING', (0, 0), (-1, -1), 4),
    ('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
    ('BOX', (0, 0), (-1, -1), 1, C_BORDER),
]))
elements.append(info_table)
elements.append(Spacer(1, 4))
elements.append(HRFlowable(width="100%", thickness=1.5,
color=C_MED_GREEN, spaceBefore=2, spaceAfter=4))

# PART A. DATA
elements.append(Paragraph("PART A. Data", h1_style))
elements.append(HRFlowable(width="100%", thickness=1,

color=C_ACCENT_MINT, spaceBefore=1, spaceAfter=4))
data_info_text = ""
<b>Source & Full URL:</b> World Bank Open Data (World Development
Indicators)<br/>
<b>URL:</b> https://data.worldbank.org/country/philippines<br/>
<b>Description:</b> Real GDP vs Population of the Philippines (2005-
2019).<br/>
<b>Variables:</b> Independent (<i>x</i>) = Population (Millions) |
Dependent (<i>y</i>) = Real GDP (Billions USD, Constant 2015)
"""
elements.append(Paragraph(data_info_text, body_style))

obs_headers = [Paragraph("Year", table_header_style),
Paragraph("Population (x) [Millions]", table_header_style),
Paragraph("Real GDP (y) [Billions USD]", table_header_style)]
obs_rows = [obs_headers]
years = list(range(2005, 2020))
for i in range(n):
    obs_rows.append([str(years[i]), f"{x[i]:.1f}", f"{y[i]:.1f}"])

obs_table = Table(obs_rows, colWidths=[100, 220, 220])
obs_table.setStyle(TableStyle([
    ('BACKGROUND', (0, 0), (-1, 0), C_DARK_GREEN),
    ('ALIGN', (0, 0), (-1, -1), 'CENTER'),
    ('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
    ('FONTSIZE', (0, 0), (-1, -1), 7.5),
    ('BOTTOMPADDING', (0, 0), (-1, -1), 2),
    ('TOPPADDING', (0, 0), (-1, -1), 2),
    ('GRID', (0, 0), (-1, -1), 0.5, C_BORDER),
    ('ROWBACKGROUNDS', (0, 1), (-1, -1), [colors.white,
C_PASTEL_BG])
]))
elements.append(obs_table)
elements.append(Spacer(1, 6))

# PART B. PYTHON SCRIPT
elements.append(Paragraph("PART B. Python Script", h1_style))
elements.append(HRFlowable(width="100%", thickness=1,
color=C_ACCENT_MINT, spaceBefore=1, spaceAfter=4))

try:
    current_script_code =
inspect.getsource(inspect.getmodule(inspect.currentframe()))
except Exception:
    current_script_code = "# Script source execution text embedded
automatically."

code_snippet = current_script_code[:1400] + "\n\n# ... [Script Code
Truncated for Page Layout] ..."
elements.append(Preformatted(code_snippet, code_style))
elements.append(Spacer(1, 6))

# PART C. REGRESSION RESULTS
elements.append(Paragraph("PART C. Regression Results", h1_style))
elements.append(HRFlowable(width="100%", thickness=1,
color=C_ACCENT_MINT, spaceBefore=1, spaceAfter=4))

card_1 = [[Paragraph("01", card_num_style)], [Paragraph("MODEL FIT",
card_title_style)], [Paragraph(f"<b>y = {a0:.2f} + {a1:.2f}x</b>",
card_text_style)]]
card_2 = [[Paragraph("02", card_num_style)], [Paragraph("ACCURACY",
card_title_style)], [Paragraph(f"<b>r^2 = {r2:.4f}</b>",
card_text_style)]]
card_3 = [[Paragraph("03", card_num_style)], [Paragraph("ERROR",
card_title_style)], [Paragraph(f"<b>s<sub>y</sub>/x</sub> =
{sy_x:.2f}</b>", card_text_style)]]
card_4 = [[Paragraph("04", card_num_style)],
[Paragraph("PREDICTION", card_title_style)],
[Paragraph(f"<b>[y_predicted:.1f]B USD</b>", card_text_style)]]

def make_card_table(content):
    t = Table(content, colWidths=[125])
    t.setStyle(TableStyle([
        ('BACKGROUND', (0, 0), (0, 0), C_DARK_GREEN),
        ('BACKGROUND', (0, 1), (0, 1), C_MED_GREEN),
        ('BACKGROUND', (0, 2), (0, 2), C_CARD_BG),
        ('ALIGN', (0, 0), (-1, -1), 'CENTER'),
        ('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
        ('TOPPADDING', (0, 0), (-1, -1), 2),
        ('BOTTOMPADDING', (0, 0), (-1, -1), 2),
        ('BOX', (0, 0), (-1, -1), 1, C_BORDER),
    ]))
    return t

cards_table = Table([make_card_table(card_1),
make_card_table(card_2), make_card_table(card_3),
make_card_table(card_4)], colWidths=[132, 132, 132])
cards_table.setStyle(TableStyle([('LEFTPADDING', (0, 0), (-1, -1),
0), ('RIGHTPADDING', (0, 0), (-1, -1), 0)]))
elements.append(cards_table)
elements.append(Spacer(1, 4))

# 1. Regression Equation & Parameters
elements.append(Paragraph("1. Regression Equation & Calculated
Metrics", h2_style))
metrics_table_data = [
    [Paragraph("Metric", table_header_style), Paragraph("Formula /
Notation", table_header_style), Paragraph("Calculated Value",
table_header_style)],
    ["Fitted Regression Model", "y = a0 + a1 * x", f"y = {a0:.4f} +

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```
{a1:.4f}x"]],
    ["Intercept (a0)", "a0 = y_bar - a1 * x_bar", f"{a0:.4f} Billion USD"],
    ["Slope (a1)", "a1 = (n*Σxy - ΣxΣy) / (n*Σx² - (Σx)²)", f"{a1:.4f} Billion USD / Million people"],
    ["Sum of Squared Residuals (SSE)", "Sr = Σ(y - y_pred)²", f"{Sr:.4f}"],
    ["Coefficient of Determination", "r² = (St - Sr) / St", f"{r2:.4f}"],
    ["Standard Error of Estimate", "s_y/x = sqrt(Sr / (n - 2))", f"{sy_x:.4f} Billion USD"]
]
```

```
table = Table(metrics_table_data, colWidths=[165, 185, 190])
table.setStyle(TableStyle([
    ('BACKGROUND', (0, 0), (-1, 0), C_DARK_GREEN),
    ('TEXTCOLOR', (0, 0), (-1, 0), colors.white),
    ('FONTSIZE', (0, 0), (-1, -1), 7.5),
    ('BOTTOMPADDING', (0, 0), (-1, -1), 2),
    ('TOPPADDING', (0, 0), (-1, -1), 2),
    ('GRID', (0, 0), (-1, -1), 0.5, C_BORDER),
    ('ROWBACKGROUNDS', (0, 1), (-1, -1), [colors.white, C_PASTEL_BG])
]))
elements.append(table)
elements.append(Spacer(1, 4))
```

```
# 2. Plots
elements.append(Paragraph("2. Graph with Fitted Line & Residual Plot", h2_style))
img_table = Table([[Image('fit_plot.png', width=260, height=130), Image('residual_plot.png', width=260, height=130)], colWidths=[270, 270])
img_table.setStyle(TableStyle([('LEFTPADDING', (0, 0), (-1, -1), 0), ('RIGHTPADDING', (0, 0), (-1, -1), 0)]))
elements.append(img_table)
elements.append(Spacer(1, 4))
```

```
# 3. Interpretation of Results
elements.append(Paragraph("3. Interpretation of Results", h2_style))
elements.append(Paragraph("The simple linear regression model indicates a strong positive linear relationship between population scale and Real GDP in the Philippines.", body_style))
```

```
bullet_interpretation = f"""
• <b>Slope (a1 = {a1:.4f}):</b> An increase of 1 million people
```

```
correlates to <b>${a1:.4f} billion USD</b> in Real GDP.<br/>
• <b>Intercept (a0 = {a0:.4f}):</b> Acts as a linear baseline offset parameter.<br/>
• <b>Coefficient of Determination (r² = {r2:.4f}):</b> <b>(r2 * 100:.2f)%</b> of variance in GDP is explained by population size.<br/>
• <b>Standard Error (s_y/x = {sy_x:.4f}):</b> Predictions vary by an average of only <b>${sy_x:.4f} billion USD</b>.<br/>
• <b>Residuals Analysis:</b> Balanced $\pm 5$ billion USD variance around 0 confirms linear consistency.
"""
elements.append(Paragraph(bullet_interpretation, body_style))
elements.append(Spacer(1, 4))
```

```
# 4. Out-of-Sample Prediction (Placed after Interpretation)
elements.append(Paragraph("4. Out-of-Sample Prediction", h2_style))
```

```
pred_box_data = [[
    Paragraph(
        f"<b>Target Population (x):</b> {x_target:.1f} Million | <b>Predicted GDP (y):</b> <b>{y_predicted:.4f} Billion USD</b><br/>"
        f"<b>Substitution:</b> y = {a0:.4f} + ({a1:.4f} * {x_target:.1f})<br/>"
        f"<b>Significance:</b> Provides baseline macroeconomic forecasting for public infrastructure planning.",
        body_style
    )
]]
pred_table = Table(pred_box_data, colWidths=[540])
pred_table.setStyle(TableStyle([
    ('BACKGROUND', (0, 0), (-1, -1), C_PASTEL_BG),
    ('BOX', (0, 0), (-1, -1), 1, C_ACCENT_MINT),
    ('PADDING', (0, 0), (-1, -1), 4),
]))
elements.append(pred_table)
```

```
doc.build(elements)

if os.path.exists('fit_plot.png'):
    os.remove('fit_plot.png')
if os.path.exists('residual_plot.png'):
    os.remove('residual_plot.png')

print(f"Report successfully compiled to '{pdf_file}'.")
```

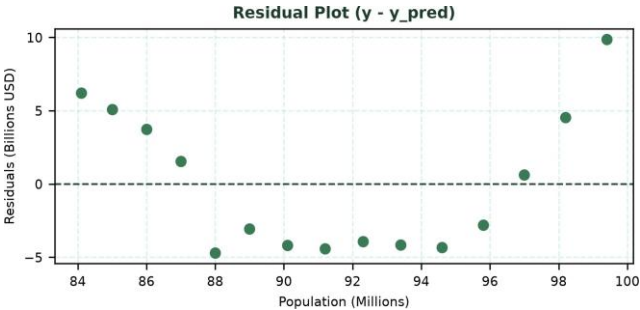
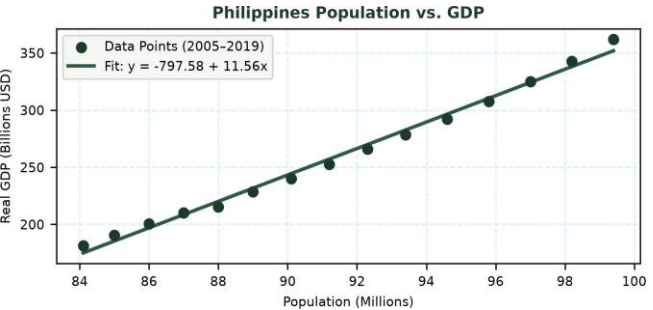
PART C. Regression Results

01	02	03	04
MODEL FIT	ACCURACY	ERROR	PREDICTION
y = -797.58 + 11.56x	r² = 0.9928	s _{y/x} = 5.01	416.7B USD

1. Regression Equation & Calculated Metrics

Metric	Formula / Notation	Calculated Value
Fitted Regression Model	y = a0 + a1 * x	y = -797.5812 + 11.5646x
Intercept (a0)	a0 = y_bar - a1 * x_bar	-797.5812 Billion USD
Slope (a1)	a1 = (n*Σxy - ΣxΣy) / (n*Σx² - (Σx)²)	11.5646 Billion USD / Million people
Sum of Squared Residuals (SSE)	Sr = Σ(y - y_pred)²	326.6997
Coefficient of Determination	r² = (St - Sr) / St	0.9928
Standard Error of Estimate	s_y/x = sqrt(Sr / (n - 2))	5.0131 Billion USD

2. Graph with Fitted Line & Residual Plot



3. Interpretation of Results

The simple linear regression model indicates a strong positive linear relationship between population scale and Real GDP in the Philippines.

- **Slope ($a_1 = 11.5646$):** An increase of 1 million people correlates to **\$11.5646 billion USD** in Real GDP.
- **Intercept ($a_0 = -797.5812$):** Acts as a linear baseline offset parameter.
- **Coefficient of Determination ($r^2 = 0.9928$):** **99.28%** of variance in GDP is explained by population size.
- **Standard Error ($s_{y/x} = 5.0131$):** Predictions vary by an average of only **\$5.0131 billion USD**.
- **Residuals Analysis:** Balanced ± 5 billion USD variance around 0 confirms linear consistency.

4. Out-of-Sample Prediction

Target Population (x): 105.0 Million | **Predicted GDP (y): 416.7014 Billion USD**

Substitution: $y = -797.5812 + (11.5646 * 105.0)$

Significance: Provides baseline macroeconomic forecasting for public infrastructure planning.

